Real-Time 2.2-Gb/s Water-Air OFDM-OWC System With Low-Complexity Transmitter-Side DSP

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***Abstract*—Underwater optical wireless communication (UOWC) is of great interest to the academic and the industry community. In this article, we propose a low-complexity and effective joint transmitter-side digital signal processing (DSP) including geometric shaping, time-domain tone reservation (TR), and clipping. The peak-to-average power ratio reduction performance and implementation complexity of the proposed time-domain TR are extensively analyzed. We then develop a real-time 2.2-Gb/s system, which is optimized for its shaping ratio and clipping ratio. With the help of the efficient DSPs, an 8-dB received optical power enhancement is realized under the bit-error-rate threshold of 3.8×10^{-3}. We successfully demonstrate a time-multiplexed four 4K video transmission in real-time using the proposed scheme over a 3.6-m underwater and 8-m free-space channel. The implementation details, as well as the analyses of system stability, resource utilization, and latency, are presented. The results validate the feasibility and effectiveness of the proposed scheme and a 2.2-Gbit/s real-time OWC system is demonstrated for a water-air communication link.**

***Index Terms*—OFDM, PAPR, real-time FPGA, Underwater optical wireless communications.**

I. Introduction

Underwater activities expedite the growth in various applications, such as sea-floor resource mining, offshore explorations, underwater salvage, pollution monitoring and climate change monitoring using underwater internet of things (UIoT). Remotely controlled vehicles (ROVs) and autonomous underwater vehicles (AUVs) are used to cover a much larger surveillance area for the aforementioned applications. To provide rich and timely surveillance information, real-time high-definition (HD) video transmissions would be desirable [1], leading to the surge of communication data rate. Traditional acoustic communication can support long-range transmission, however, it has very limited bandwidth and suffers from large latency due to its low transmission speed (~1.5 km/s). For the underwater link, the data rate of the surveillance video transmission would be far exceeding the transmission capacity that acoustic communication can offer. Using a cable (termed as "umbilical") is an alternative solution to enable the collection of surveillance data and the provision of real-time control and electric power for underwater ROVs or robotic modules. However, the umbilical inherently limits mobility, thus not suitable for devices that require frequent motion and deep diving. For UIoT sensors, wired communication is not a cost-efficient solution, either.

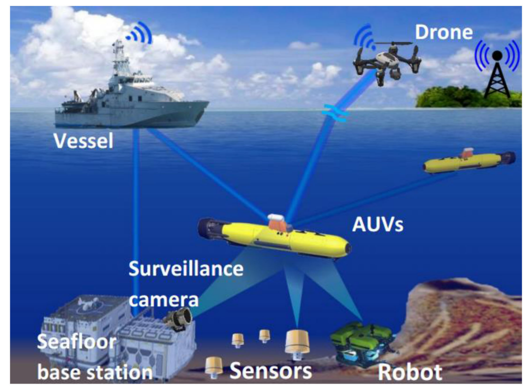

Fig. 1. Illustration of multiple access underwater and water-air OWC.

Optical wireless communication (OWC) using a blue laser diode (LD) has been demonstrated for high-speed free-space transmission [2], [3] and for systems with terminals' mobility [4], [5]. Moreover, blue-to-green light exhibits a relatively low loss in water, resulting in a long available transmission distance, possibly of around a hundred meters [6]. Therefore, underwater OWC (UOWC) is the preferred choice as it provides significantly higher data rates than traditional solutions and has high flexibility. To provide maximum flexibility, the captured HD surveillance video can be beamed to a drone above the water to relay the signal back to a remote land site control center through a low-loss RF wireless link. Compared with the RF signal and acoustic signal, the optical signal exhibits lower loss through the water-air interface and relatively small attenuation for both air and water links [7]. The possible scenarios of future underwater and water-air OWC are illustrated in Fig. 1.

Recently, quite a few UOWC schemes have been demonstrated for high-speed underwater and air-water links, respectively. It has been shown that the blue LD enables high-speed

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UOWC [8], and 12.62-Gbit/s transmission over 35 meters with a 450-nm LD [9] has been achieved. For the long-range transmission, a 100-m/500-Mbit/s UOWC with a 520-nm green LD using return-to-zero on-off-keying (NRZ-OOK) is demonstrated [10]. In [11], a 5.5-Gbit/s air-water OWC system based on offline orthogonal frequency division multiplexing (OFDM) is demonstrated over 26 m. High-speed 30 Gb/s UOWC using PAM4 is achieved by injection locking [12], and in [13] a 256Gb/s SDM-based PAM4 air-water OWC system is investigated. For real-time systems, a 25-Mbit/s NRZ-OOK UOWC system is implemented [14] and a 10-Mbit/s UOWC system is verified in sea-trial over a 10-meter bi-directional link [15]. Among these researches, high data rate is usually achieved using high-spectrum-efficiency OFDM. Despite its advantages, one major drawback of the OFDM system is the high peak-to-average power ratio (PAPR) that leads to power inefficiency and signal distortion due to the non-linearity of various devices in the system [16]. This problem restricts the transmitted power, increases the quantization errors, and degrades system performance. Thus, it imposes a trade-off between the signal power and the signal distortion.

To alleviate the issue, firstly, signal pre-distortion by geometric shaping (GS) can be used to compensate the nonlinearity at the transmitter side [17]. However, the special-shaped constellation, such as circular 8-ary quadrature amplitude modulation (8QAM) [18], requires a complex de-mapping scheme. By contrast, the GS of standard rectangular QAM exhibits lower complexity in demapping, which is preferable for real-time implementation. Secondly, different approaches have been adopted to alleviate the high PAPR issue in multicarrier OWC systems. These include discrete Fourier transform (DFT)-spread precoding [19], [20] and selected mapping (SLM) [21]. However, both SLM and DFT-spread pre-coding are of high computational complexity, due to the iterations and the parallel variable-length DFT design, respectively. Thus, they are not suitable for high-speed real-time implementation. The tone reservation (TR) method [22] is shown as an effective method to improve the PAPR performance of OFDM systems, while at the expense of significantly increased complexity and latency due to the iterative procedures and several-fold inverse fast Fourier transform (IFFT) operations.

In this paper, we present extensive analyses and results for our recent proposed scheme with a GS-8QAM mapping scheme and a joint PAPR reduction via a simplified time-domain TR combined with clipping (TRC) over a water-air OWC system [23]. A field-programmable gate array (FPGA)-based real-time 2.2-Gb/s OWC system over 3.6-m underwater and 8-m free-space link to support multiple HD video channels is demonstrated. The PAPR reduction performance and implementation complexity of the proposed time-domain TR are analyzed. In the experiment, the proposed scheme shows a significant 8-dB improvement in received optical power (ROP) sensitivity under the bit-error-rate (BER) threshold of 3.8×10^{-3}. Furthermore, the scheme's stability has been validated by real-time video-data transmission, with a small BER variance and no video packet loss. We successfully demonstrate a time-multiplexed four 4K video transmission over the real-time OWC system, aided by the proposed scheme and a Reed-Solomon (RS) forward error correction (FEC). Resource utilization and system latency are also studied in the real-time system.

The rest of this paper is organized as follows. Section II gives the detailed principles of the proposed low-complexity transmitter-side DSP. In session III, we describe the experimental setup and hardware implementation of the water-air OWC system. Section IV presents the experimental results and discussions. Finally, Section V concludes this paper.

II. DSP Principles

A. Geometric Shaping (GS)

Geometric shaping has been studied for optical fiber communication to realize transmission that approaches the channel capacity. Typically, the constellation will be shaped to circular constellation [24]. However, the spatially shaped constellation requires high computational complexity for the demapping process. For a standard rectangular 8QAM constellation set $\{\pm 1 \pm \mathrm{i}, \pm 3 \pm \mathrm{i}\}$, the points are equally spaced with the hard decision lined in the middle of two nearest neighbors. However, under the effect of non-linearity, the constellation spaces will no longer be equidistant, with the outer symbols exhibiting larger distortion that results in a higher symbol error rate [25].

In this paper, to mitigate the nonlinearity effect, we propose to use a geometrically shaped 8QAM set $\{\pm p_1 \pm p_1 \mathrm{i}, \pm p_2 \pm p_1 \mathrm{i}\}$, where $3{p_1}^2 + {p_2}^2 = 12$, to keep a constant power as the standard 8QAM, and p_2/p_1 is defined as a shaping ratio ρ and should be optimized when in use. Note that, this GS-8QAM scheme is different from specially shaped constellation such as circular 8QAM [18], which required a fine-designed decision threshold or K-means clustering [26]. The corresponding demapping method of our scheme can be based on the traditional decision threshold of the standard rectangular 8QAM to maintain low real-time implementation complexity. Albeit only rectangular 8QAM signal is experimentally demonstrated in this paper, the scheme can be applied to higher-order QAM signal with modification on the shaping ratio optimization. Taking 16QAM as an example, the constellation set to be optimized is $\{\pm p_1 \pm p_1 \mathrm{i}, \pm p_2 \pm p_1 \mathrm{i}, \pm p_1 \pm p_2 \mathrm{i}, \pm p_2 \pm p_2 \mathrm{i}\}$ with a shaping ratio of p_2/p_1. For further higher-order QAMs, the scheme is still applicable but requires the optimization of more than one parameter due to the increase of the constellation points.

B. Time-Domain Tone Reduction (TR)

The TR method is proposed to reduce the PAPR of multi-carrier signals by reserving a small number of non-data-bearing subcarriers as peak reduction tones (PRTs) [27]. For OFDM signals, as the subcarriers are orthogonal, it can effectively reduce the probability of high peak power without affecting the signals loaded on the data-bearing subcarriers. Thus, they can be easily separated at the receiver without additional side information, as the data and the PRTs are mutually exclusive. The main disadvantage of the method is that it may sacrifice some subcarriers and consequently reduce spectrum efficiency. Thus, this method is not favorable for the RF wireless system

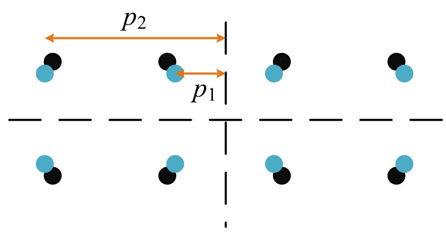

Fig. 2. GS rectangular 8QAM constellation.

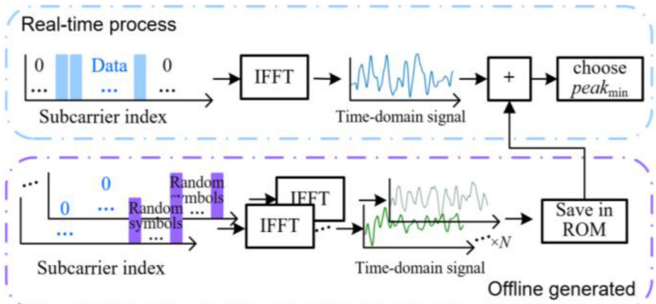

Fig. 3. Block diagrams of the proposed time-domain TR-based PAPR reduction scheme.

with precious spectrum resources. While in the case of a typical OWC system, some edge subcarriers exhibit very low SNRs due to the limited bandwidth of the optoelectronic devices. Thus, considering a given sample rate hardware, i.e. digital-to-analog converter (DAC), a small number of edge subcarriers can be exploited for PAPR reduction by TR method without affecting the signal loaded within the usable bandwidth. The objective of the method is to choose proper frequency-domain variables to minimize the peak value of the time-domain signal. This minimax optimization problem is theoretically analyzed in [22], which is convex on the set, and a linear programming model can be applied. However, the conventional TR scheme is based on frequency-domain variables to minimize the peak value of the time-domain signal. The approach requires iterative procedures, even for some simplified methods, such as the time-domain kernel matrix (TKM-TR) [28] scheme. Thus, it will lead to high computational complexity and long convergence time.

In our recent paper [23], we develop a new low-complexity time-domain TR scheme for the real-time OFDM system implementation. The main idea is to add a data-dependent time-domain sequence, termed as peak reduction sequence (PRS), to the original OFDM signal after frequency-time domain transformation to reduce its peaks, as illustrated in Fig. 2. Each PRS is offline generated by modulating different random data on the unused subcarriers. In the transmitter, when the real-time mapped data come in, we allocate the data on the subcarriers with good frequency response, whereas the subcarriers with low SNRs are zero-padded. After the IFFT, the generated time-domain signal is added with the pre-stored PRSs and the one with the minimum peak will be chosen for transmission.

The PAPR of one OFDM block can be expressed as:

$$PAPR = \frac{\max |x(n)|^2}{\mathrm{E}\left[|x(n)|^2\right]}, \tag{1}$$

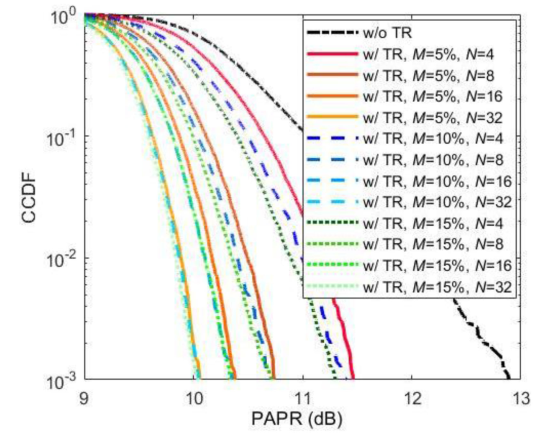

Fig. 4. CCDF of PAPR with/without time-domain TR using different number of peak reduction tones and different number of pre-stored PRSs. The PRSs are generated on subcarriers that occupy a percentage of $M = 5\%$, 10% and 15% of data subcarriers. PRSs set size $N = 4$, 8, 16, and 32.

where E[·] denotes the expectation operation. PAPR is evaluated by the complementary cumulative distribution function (CCDF) defined as the probability of PAPR exceeding a certain threshold $PAPR_0$:

$$CCDF(PAPR_0) = \Pr(PAPR > PAPR_0) \tag{2}$$

where $PAPR_0$ is any positive value and Pr[·] is the probability function.

We then optimize the TR scheme with two parameters considered: the ratio of the peak-reduction tone number over the data-bearing carrier number and the number of pre-stored PRSs, represented by M and N, respectively. The original OFDM symbols are generated using a 1024-point IFFT, with 315 subcarriers carrying data. For real implementation, the number of data-bearing subcarriers will be determined by the frequency response of the system. N pre-stored PRSs are generated by modulating random data on M unused subcarriers. The remaining subcarriers are zero-padded. Fig. 4 shows the PAPR performance comparison of 8QAM-OFDM symbols with and without using the time-domain TR method. It is seen that with the increase in pre-stored PRSs set size N, the PAPR performance of OFDM signals using time-domain TR is improved. However, the improvement diminishes with the increase in N. Moreover, we can see that the number of peak reduction tones will also affect the PAPR performance, albeit less significant compared to the PRSs set size. Besides, the influence is becoming negligible with N further increasing. This indicates that the randomness of the PRSs is sufficient when the PRSs set size is large enough.

However, there is a trade-off between performance versus complexity. The complexity of the time-domain TR is summarized as follows: the required addition operations are NL, the peak search computational complexity is $O(L)$, and the minimum search requires $O(N)$. Whereas, the prior work [22] normally requires several IFFT/FFTs, with a complexity of $O(KN\log N)$, where K is the iteration number. The required numbers of real multiplications, divisions, and additions are summarized in Table I, with the comparison of some PAPR reduction approaches

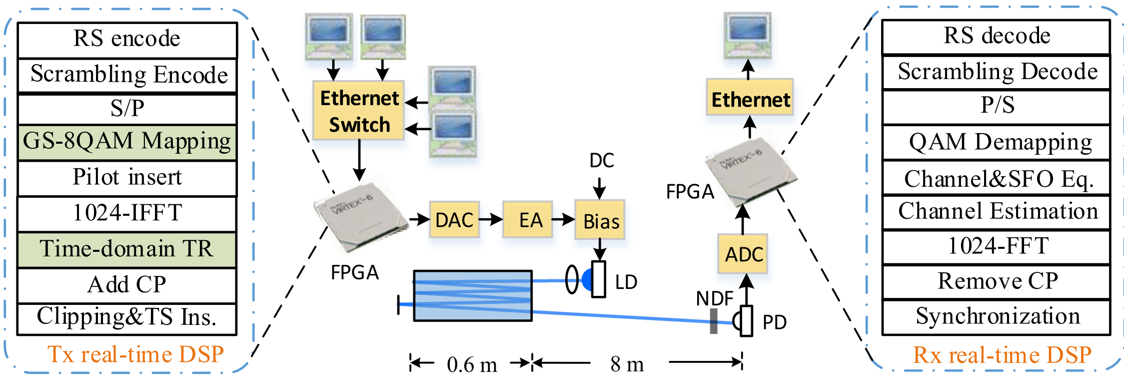

Fig. 5. The real-time DSP flowchart and experimental setup of the OFDM-based OWC system. (DC: direct current; TS ins.: TS insertion; channel & SFO Eq.: channel estimation and sampling frequency offset equalization.)

TABLE I
TRANSMITTER-SIDE COMPLEXITY COMPARISON

Scheme	Real Multiplication	Real Division	Real Addition
TKM-TR [28]	$K(3M+2NM)$	$K(2M)$	N/A
DFT [20]	$3N_d^2$	N/A	$2N_d(6N_d-1)$
This work	N/A	N/A	LN

(M: the length of symbol set; K: iteration times, N_d: DFT-precoding matrix size.)

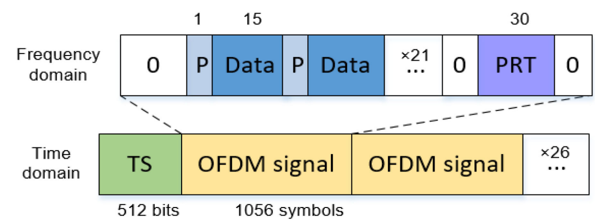

Fig. 6. The packet structure in the frequency domain and time domain. (P: pilot; TS: training sequence.)

including a lower complexity TKM-TR scheme [28] and a DFT-spread-OFDM [20]. Note that the additions are omitted for the first one. For the latter one, the total computational complexity composes of not only an additional N_d-points DFT block in the transmitter side but also the IDFT operation with comparable complexity in the receiver side, where N_d is the DFT-precoding matrix size.

As the complexity and the required memory resource increases linearly with *N*, to balance the complexity and performance, eight PRSs occupying 9% of data-carrying subcarriers are used for PAPR reduction in the following experiment. To further improve the performance, the clipping method [27] is introduced. The clipping ratio r is defined as the maximum permissible amplitude over the mean power. Note that clipping will lead to a trade-off between signal power and signal distortion, thus, the clipping ratio needs to be optimized to achieve the optimal performance.

III. Experimental Setup and Implementation

In this section, the experimental setup and the detailed hardware implementation of the real-time OWC system will be presented. The corresponding DSP flowchart with GS and TRC is illustrated in Fig. 5. The real-time OFDM transmitter and receiver were implemented in two FPGA platforms: Xilinx ML605 equipped with a 14-bit DAC (AD9739A) and Xilinx VC707 equipped with a 10-bit analog-to-digital converter (ADC, EV10AQ190), respectively. The bitstream of the video was generated by a PC and sent through a 1-Gbps Ethernet port to the FPGA. The bitstream was first scrambled to prevent long 1s or 0s of the video data to ensure the randomness of the data. Then, after the GS-8QAM mapping, the signal was modulated on 315 parallel subcarriers, starting from the 17-th subcarrier. Meanwhile, the pilot symbols were inserted in every 16 subcarriers to facilitate the linear-interpolated channel estimation and sampling frequency offset equalization [29]. Note that the pilot is specially designed using only binary phase-shift keying (BPSK) signals, i.e. $\{\pm 1\}$, to reduce the complexity for channel estimation. This is owing to the neglectable division operations at the receiver accordingly. Then, the time-domain signal was then generated by a 1024-point IFFT and the real part was taken. This process is equivalent to the Hermitian symmetry [30]. Then the time-domain TR using subcarriers 461–490 was performed to avoid the high peak of the signal. The TR scheme was performed to reduce signal PAPR. After time-domain signal addition, the PAPR estimation was performed to realize the TR scheme. As the signal mean power fluctuation is usually small, the signal peak can be used as the index instead of PAPR in the real-time system for cost-saving. Next, clipping with different ratios was utilized to further reduce the PAPR. At last, for every 26 OFDM symbols, a 512-point training sequence (TS) was inserted for synchronization. It is worth noting that the TS also contained BPSK signals to reduce synchronization complexity. The frame structure is illustrated in Fig. 6, in both frequency domain and time domain. Then, the signal after scaling was output by the DAC at a sampling rate of 2.5 GSa/s. Considering the 1/32 cyclic prefix (CP) and the pilot sequence, the net data rate achieved is about 2.2 Gb/s.

After an electrical amplifier (EA, ZX60-43-S+), the signal coupled with a DC component (via a DC-bias, ZFBT-6GW+) was used to drive a blue LD (Osram, PLPB450). The amplification gain and the bias voltage were optimized at 16 dB and 4.8 V.

TABLE II
SYSTEM PARAMETERS

Parameter	Value	Parameter	Value
Modulation format	8QAM	Amplification gain	16 dB
IFFT size	1024	Bias voltage	4.8 V
Data subcarrier number	315	Transmit optical power	26 dBm
Pilot subcarrier number	21	Divergence angle	0.12°
PRT number	30	Channel loss	~7 dB
PRSs set size	8	APD sensitivity @450 nm	<5 A/W

After the collimating lens, the light, with a divergence angle of 0.12 degree, was reflected five times by mirrors over a 0.6-m water tank to form a ~3.6-m water path. Then, it passed an 8-m free-space channel and was detected by an avalanche photodiode (APD, Hamamatsu C5658, 1-GHz). The total loss of the water channel was ~7 dB, which was mainly from water attenuation and reflection loss. After being sampled by the ADC, the signal was sent to FPGA for real-time processing. The detailed system parameters are summarised in Table II.

At the receiver, synchronization was first performed using the training sequence. Then, the inverse process was performed to recover the symbols in the frequency domain. The signal loaded on the reserved tones could be simply discarded in the frequency domain. Thus, the receiver-side DSP remained the same as the conventional OFDM signal with no additional process required. Then, the simultaneous sampling frequency offset (SFO) compensation and channel equalization were performed. Here we adopted a pilot-aided and linear-interpolated channel estimation scheme proposed in [31]. The channel response can be estimated as:

$$H(i) = \frac{(i-k_1)[\tilde{H}(j_1) - \tilde{H}(k_1)]}{j_1 - k_1} + \tilde{H}(k_1), \qquad (3)$$

where j_1 is the sub-carrier index of the right adjacent pilot in the i-th data-carrying sub-carrier, k_1 is the sub-carrier index of the left adjacent pilot. $\tilde{H}(\cdot)$ is the channel response factor (CRF) of the sub-carrier estimated by the inserted pilot, and $H(i)$ is the CRF of the i-th subcarrier. It should be mentioned that if the gap between the two adjacent pilots is 2^N, the real-time process of the pilot-aided channel estimation can be implemented by only using shift register and adder-subtractor.

In order to avoid using the complex division, only phase compensation is carried out in the channel equalization, namely, only the numerator part in (3) is executed in the equalization stage. As for the denominator part that is related to amplitude compensation, it will be used to alter the corresponding decision thresholds.

The detailed structures of the real-time DSP for GS and time-domain TR are illustrated in Fig. 7(a) and (b), respectively. The constellation points for mapping was software-controllable by using an external command from the PC. For time-domain TR, eight time-domain PRSs were first saved in the read-only memory (ROM) on board. Then, the IFFT output was replicated eight copies in parallel and each of them added one PRS. After that, the signals were stored in random access memory (RAM). After one OFDM block (64 timeslots) was generated,

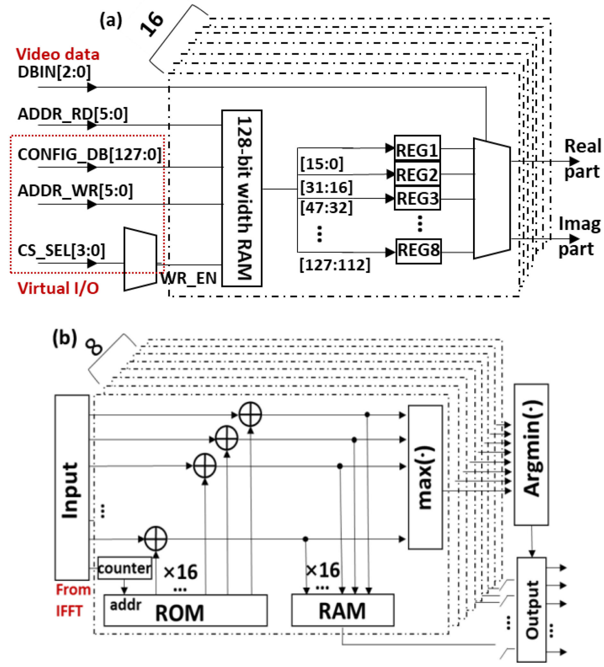

Fig. 7. The hardware structure of (a) software-defined GS, and (b) time-domain TR.

the minimum of the eight peak values was identified and the corresponding block in RAM would be selected and output. Because only the addition operation is required, the complexity of the proposed scheme can be significantly reduced.

IV. EXPERIMENTAL RESULTS AND DISCUSSIONS

In this section, the performance of the proposed GS scheme and TRC are optimized and investigated individually and jointly. A real-time four-channel 4K video transmission is demonstrated over the water-air channel, with four PCs streaming different video sources. Moreover, we emulate a real scenario using two UOWC channels from two LDs and implement the real-time TRC at the relay point. System stability, hardware resource utilization, and system latency are analyzed. At last, the system performance considering practical environment issues is discussed.

Fig. 8 shows the normalized power spectrum of the digital back-to-back system and the OWC system. The 3-dB bandwidth is around 400 MHz. The reserved subcarriers for PAPR reduction (the PRTs) are located in 1.13 GHz to 1.2 GHz, which exhibits over 20-dB fading after the system transmission, as shown in Fig. 8(b). Thus, these subcarriers cannot be exploited to transmit information resulting in no sacrifice of the spectrum efficiency.

We first optimize the shaping ratio for the GS-8QAM signal, as shown in Fig. 9. Note that the estimated average SNR over all data subcarriers is around 15.83 dB, therefore, 8QAM is chosen as the modulation format for its performance in the SNR region. From the corresponding constellation diagram of the original 8QAM in the *212*-th subcarrier, we can see that the higher-power points suffer larger distortion and more errors. Thus, by

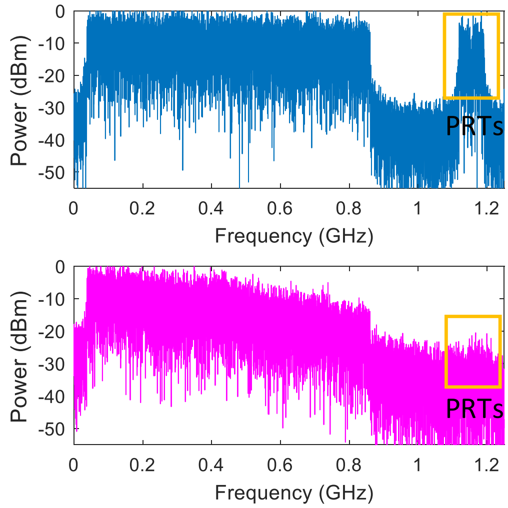

Fig. 8. The normalized power spectrum of (a) digital back-to-back system, and (b) the OWC system.

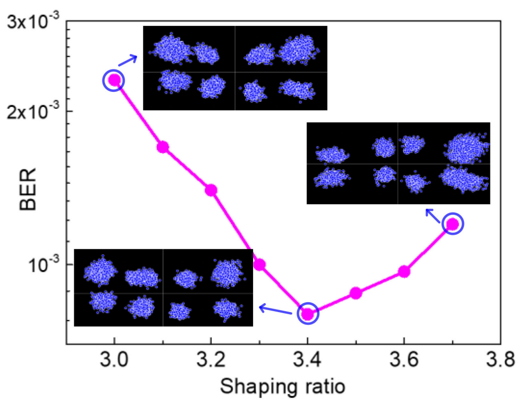

Fig. 9. BER performance of the geometric shaping ratio ρ, with the real-time corresponding constellation diagrams of the *212*-th subcarrier at $\rho = 3$ (no shaping), 3.4 (optimal), and 3.7.

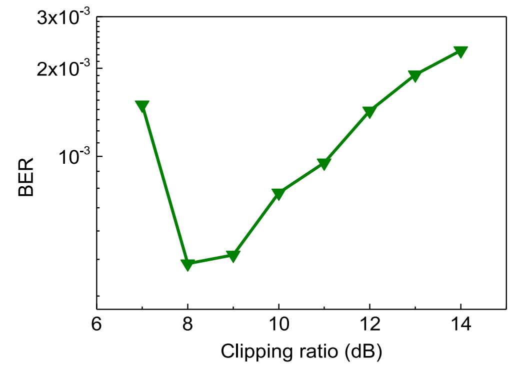

Fig. 10. BER performance of GS and clipping scheme versus the clipping ratio.

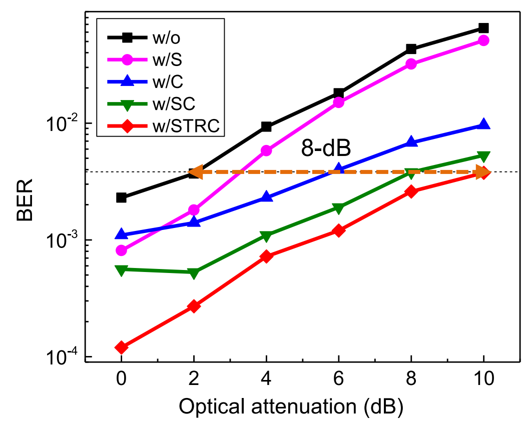

Fig. 11. BER performance versus different ROP of original OFDM signals (w/o), with GS (w/S), with clipping (w/C), with shaping and clipping (w/SC), and with the joint method (w/STRC).

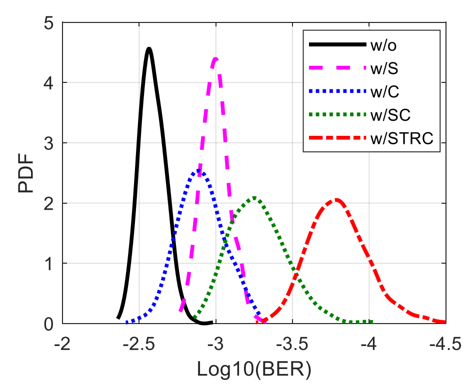

Fig. 12. Pre-FEC BER distribution of 1000 video data transmission and the corresponding variance of each scheme.

increasing the shaping ratio, the performance will be improved first. However, further increasing the outer points' power will deteriorate the inner points' performance. The optimal shaping ratio is found to be 3.4.

We then investigate the optimization of the clipping scheme, together with GS-8QAM, as shown in Fig. 10. The peak power of each block is controlled by the clipping function, and the signal will be scaled to match the maximum output of the DAC. With the clipping ratio increased, the signal power increases, which leads to a higher signal-to-noise ratio. With further reduction in the peak power of the signal, the clipping noise becomes dominant and the overall performance degrades. The optimal clipping ratio r is found to be 8 dB.

We further investigate the BER performance of different schemes versus ROP by using a variable neutral density filter (NDF). As shown in Fig. 11, all schemes provide system performance improvement in terms of reduced BER or increased receiver sensitivity. With no external optical attenuation, the proposed scheme achieves more than one-order improvement in BER when compared with the original one, whereas the GS with clipping (SC) schemes have limited improvement in the high ROP region. This mainly attributes to the pre-PAPR reduction of the TR approach that prevents high peak power of the signal and thus the clipping-caused extreme signal distortion is evaded. With the joint DSP, we achieve an ROP improvement of 8 dB under 3.8×10^{-3} hard-decision FEC (HD-FEC) BER threshold.

We then analyze the BER statistics using the raw data of multiple pre-stored videos for 10 hours to evaluate the stability of the link. We can see from Fig. 12 that clipping may cause an increase in BER variance as the random peak power leads

Fig. 13. Real-time 4K video transmission.

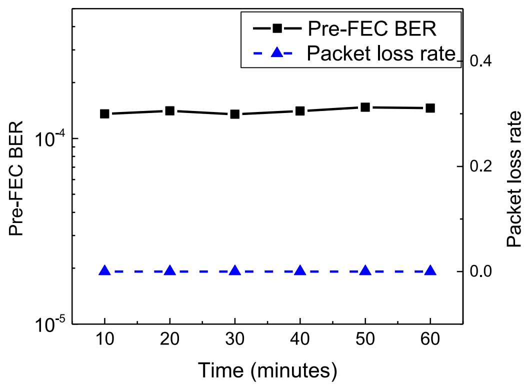

Fig. 14. Stability test over one hour using Pre-FEC BER and packet loss rate.

to different signal distortion from clipping. Combining the TR scheme can avoid a severe clipping effect. Our joint scheme exhibits the lowest variance, i.e. 5.53×10^{-9}, compared with 2.56×10^{-7} of the original OFDM scheme.

At last, we demonstrate a time-multiplexed four 4K video transmission in real-time over a water-air channel, with four PCs streaming different video sources. The 4K video stream is sliced to 1358-byte length packets based on the MPEG-TS protocol and sent through the Ethernet port to FPGA. Then the (201, 194) RS code based on the Xilinx IP Core is performed and the encoded data is modulated to OFDM symbols with the aforementioned DSPs in real-time. After the transmission, the signal is demodulated by FPGA and the decoded bits are sent back to the computer through Ethernet port. For different video streaming, the data packets are transmitted sequentially and multiplexed in time division. The experiment is implemented under normal illumination, as shown in Fig. 13. Note that the required data rate for one compressed 4K video stream is $\sim$25 Mbps, which only occupies 1.1% of the system capacity. The system stability is verified with no packet loss over a one-hour video transmission. The pre-FEC BER is also shown in Fig. 14. Note that the BER measurement is based on the pre-stored 4K video data before the FEC.

Then, the hardware resource utilization and system latency are analyzed to validate the efficiency of the proposed scheme. Table III compares the hardware resource utilization of video transmission estimated by the FPGA design tool of transmitter real-time DSP with and without the proposed as well as the receiver side. It can be seen that the proposed scheme using GS

TABLE III
RESOURCE UTILIZATION COMPARISON

Scheme	Tx		Tx with GS and TRC		Rx	
Item	Utilization	%	Utilization	%	Utilization	%
Registers	29,963	9	32,836	10	84,048	13
LUTs	30,471	20	48,361	32	91,159	30
Memory	3,912	6	6,984	11	7,535	5
RAMB36E1 /FIFO36E1s	95	22	111	26	290	28
RAMB18E1 /FIFO18E1s	69	8	69	8	9	1
DSP48E1s	173	22	173	22	542	19

(LUT: look-up table, RAM: Random-access memory, FIFO: First-In, First-Out.)

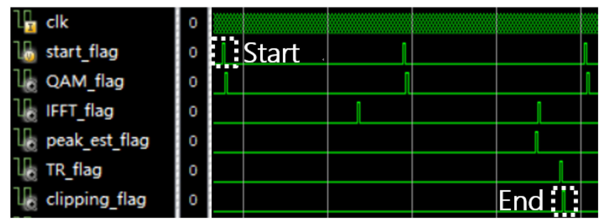

Fig. 15. Transmitter side latency analysis.

and TRC requires 50% more look-up tables (LUTs) and doubles the memory usage, compared to the conventional OFDM scheme. They are used for addition operation and buffering the signals after being added to different PRSs in TR. Whereas, no additional DSP block is required. Moreover, the receiver-side DSP maintains the same for this scheme. The results verify the feasibility and effectiveness of implementing the proposed scheme in real applications.

The latency of the transmitter-side DSP is analyzed by the timing diagram of the status flag for each DSP block, as shown in Fig. 15. The flag is asserted high to indicate that the block is finished for each OFDM symbol. The total latency is 126 clocks, while the TR requires 58. We can see that the proposed scheme introduces the system latency of 46%, which mainly attributes to the peak estimation procedure. The clock frequency is 156.25 MHz, resulting in a system latency of 0.8 µs at the transmitter side.

At last, the turbulence effect on the system performance is discussed. The turbulence will cause fluctuations in the ROP and the BER performance degradation, thus reducing the system's stability and robustness. The temporal scale of underwater optical turbulence is ms, while the duration for each data packet in this experiment (2.4×10^4 bits) is around 10 μs. In such a short time, the turbulence effect in each packet can be considered as static and causes only differences in received optical power. Our proposed scheme has demonstrated its effectiveness for different ROP, as shown in Fig. 11. Therefore, it is expected that the scheme will still be effective under turbulence. Some possible solutions to mitigate the above problem include adaptive coding schemes, tracking systems, and using beam expender [12]. Using larger beam size can improve the stability, at the expense of lower ROP. As our scheme can provide a larger power budget, it will help alleviate this issue. Besides the ROP fluctuation, the multi-path effects induced by scattering and transmission through ocean waves may cause signal pulse spreading, thus limiting the bandwidth of the OWC system. It is worthy of more comprehensive investigations of the above issues and further exploration of different mitigation schemes in a practical scenario.

V. Conclusions

In this paper, we have performed extensive simulation and experimental investigation of our proposed low-complexity and effective transmitter-side DSP scheme that combines GS-8QAM, TR and clipping methods for high-speed real-time water-air OWC link. The PAPR reduction performance and implementation complexity of the proposed time-domain TR are analyzed. To balance the trade-off between the complexity and performance, we use 9% subcarriers as peak-reduction tones and eight different time-domain sequences. We then develop a real-time 2.2-Gb/s water-air OWC system, with the optimization of the shaping ratio and clipping ratio. The joint DSP scheme provides an 8-dB ROP enhancement under the 7% HD-FEC BER threshold (3.8×10^{-3}). A time-multiplexed four 4K video transmission using the proposed scheme is successfully demonstrated in real-time over 3.6-m underwater and 8-m free-space OWC channel. The system stability is validated, and the resource utilization and latency of the system are analyzed with real-time implementation. The experimental results verify the feasibility and effectiveness of the proposed scheme and provide an OWC system for the water-air communication link.

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